

Lucas Sterzinger, PhD
Curriculum Vitae

Address removed for Public CV
<https://github.com/lsterzinger>

WORK EXPERIENCE

Mission Support/Scientific Software Developer 2023 - Present
NASA Goddard Earth Sciences Data and Information Services Center (GES DISC)
ADNET Systems, Inc

- Manage on-prem SIPS data production for AIRS (Aqua) and CrIS/ATMS (SNPP, JPSS-1, and JPSS-2) instruments
- Manage end-to-end production, ingest, and archive workflows for Earth science data
- Support transition to cloud-based environment

Graduate Student Researcher 2017 - 2023
Atmospheric Science Graduate Group, UC Davis
Dr. Adele Igel, Faculty Advisor

- Worked on research related to cloud and precipitation physics. Projects included:
 - Effect of ice crystal habit (shape) on orographic snowfall in the Sierra Nevada Mountains. (Funding: Internal)
 - Examining the relationship between mixed-phase Arctic cloud dissipation and aerosol properties. (Funding: DOE ASR; A. Igel, PI)
 - Assessing relative impacts on aerosol contained within the boundary layer and free troposphere on the microphysics and other properties of Arctic mixed-phase clouds. (Funding: DOE ASR; A. Igel, PI)

Intern Summer 2021
Summer Internship in Parallel Computational Science (SIParCS)
National Center for Atmospheric Research (NCAR), Boulder, CO

- Worked with Chelle Gentemann (Farallon Inst./NASA TOPS), Kevin Paul (NCAR), Julia Kent (NCAR), Rich Signell (USGS) and Martin Durant (Anaconda Inc.) on the development of the Kerchunk software library and its applicability and performance accessing cloud-hosted NOAA/NASA satellite data.
- Wrote documentation, blog posts, and example code on how to get started using Kerchunk - published open-source on GitHub.

Undergraduate Research Assistant 2016 - 2017
Dept. of Atmospheric Sciences, University of N. Dakota
Dr. Gretchen Mullendore, Faculty Advisor

- Worked on the “Big Weather Web” project examining potential uses for cloud computing infrastructure for numerical weather prediction.

Undergraduate Teaching Assistant 2015 - 2017
Dept. of Atmospheric Sciences, University of N. Dakota

- Independently taught Introduction to Meteorology lab, complete with weekly lectures and laboratory experiments.

Technical Support Specialist

2012 - 2017

Univ. of N. Dakota School of Medicine and Health Sciences

- Responsible for direct technology support to faculty, staff, and students. Also worked on managing video conference systems, networks, and servers.

TECHNICAL

Languages & Software: Python, Perl, Julia, Fortran

Operating Systems: Unix/Linux, MacOS, Windows

Software Packages:

- [PyRAMS](#) (maintainer) - Package for working with RAMS model data
- [Kerchunk](#) (contributor) - Cloud performant access to NetCDF4 data

EDUCATION

PhD, Atmospheric Science

2017 - 2023

University of California, Davis, Davis, CA

Dissertation: *Ice, Liquid, and Aerosol: Mixed-Phase Cloud Properties and Processes in Regional and Large Eddy Simulations*

Bachelor of Science, Atmospheric Sciences

2012 - 2017

University of North Dakota, Grand Forks, ND

Minor: Mathematics

Bachelor of Science, Aeronautics

2012 - 2017

University of North Dakota, Grand Forks, ND

PUBLICATIONS

Producing Earth science data for impact: Improved commercial cloud usability of archive model data

2026

McNally, A., Sterzinger, L. J., & Carrol, I.

Environmental Modelling & Software

<https://doi.org/10.1016/j.envsoft.2025.106846>

Above-cloud concentrations of cloud condensation nuclei help to sustain some Arctic low-level clouds

2024

Sterzinger, L. J. & Igel, A. L.

Atmospheric Chemistry and Physics

<https://doi.org/10.5194/acp-24-3529-2024>

Do Arctic mixed-phase clouds sometimes dissipate due to insufficient aerosol? Evidence from comparisons between observations and idealized simulations

2022

Sterzinger, L. J., Sedlar, J., Guy, H., Neely III, R., & Igel, A. L.

Atmospheric Chemistry and Physics

<https://doi.org/10.5194/acp-22-8973-2022>

The Effects of Ice Habit on Simulated Orographic Snowfall

2021

Sterzinger, L. J., & Igel, A. L. - *Journal of Hydrometeorology*

<https://doi.org/10.1175/JHM-D-20-0253.1>

***Models in the Cloud: A Cost Exploration
of Cloud Computing for the Atmospheric Sciences***

2017

News@Unidata Blog

<https://www.unidata.ucar.edu/blogs/news/entry/models-in-the-cloud-a>

SERVICE

UC Davis Graduate Student Association

- General Assembly Representative
- Elections Committee

2019-2022

2019-2020

UC Davis Academic Senate
Committee on Information Technology
Graduate Student Representative

2020-2021

SELECTED CONFERENCE PRESENTATIONS

Open Science Success Stories

Dec. 2022

Session Co-Convener
American Geophysics Union Fall Meeting 2022 - Chicago, IL

Arctic Mixed-Phase Clouds Sometimes Dissipate Due to Insufficient Aerosol - Evidence from Idealized Large Eddy Simulations

May 2022

Oral Presentation
European Geosciences Union General Assembly 2022 - Vienna, Austria

Arctic Mixed-Phase Clouds Sometimes Dissipate due to Insufficient Aerosol: Evidence from Idealized Large Eddy Simulations

Apr. 2022

Oral Presentation
2nd QuIESCENT Workshop - Tromsø, Norway

Fake it until you make it — Reading GOES NetCDF4 data on AWS S3 as Zarr for rapid data access

Jan. 2022

Oral Presentation and Interactive Workshop
2022 ESIP January Meeting

Cloud-performant reading of NetCDF4/HDF5/Grib2 using the Zarr library

Dec. 2021

Oral Presentation
American Geophysics Union Fall Meeting 2021

Effects of Aerosol Concentration on the Dissipation of Arctic Mixed-Phase Clouds

Dec. 2020

eLightning
American Geophysics Union Fall Meeting 2020

MEMBERSHIPS

American Meteorological Society
American Geophysics Union
European Geosciences Union

LANGUAGES

English
French (Bilingual Fluency)